Risky Income, Risky Families: Marriage and Divorce in a Volatile Labor Market

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Motivation

- Since 1970: large increases in idiosyncratic labor income risk.
- Changes in the American family:
  - Decline and delay in marriage.
  - Divorce since 1980:
    - Overall decline.
    - Rise among the elderly.
- How much of these changes can volatility explain?
  - Use a model with endogenous marriage, divorce, & fertility.

Idea

- Marriage and risk:
  - Kids are a reason for marriage- consumption commitment.
  - Marriage provide spousal insurance.

- When volatility increases, we expect:
  - Less marriage because of commitments, more due to insurance.
    - Quantitatively, less marriage.
  - Less divorce because of spousal insurance.
  - More divorce among the elderly because...
    - ... face less income risk (no use for spousal insurance).
    - ... don't have kids anymore.

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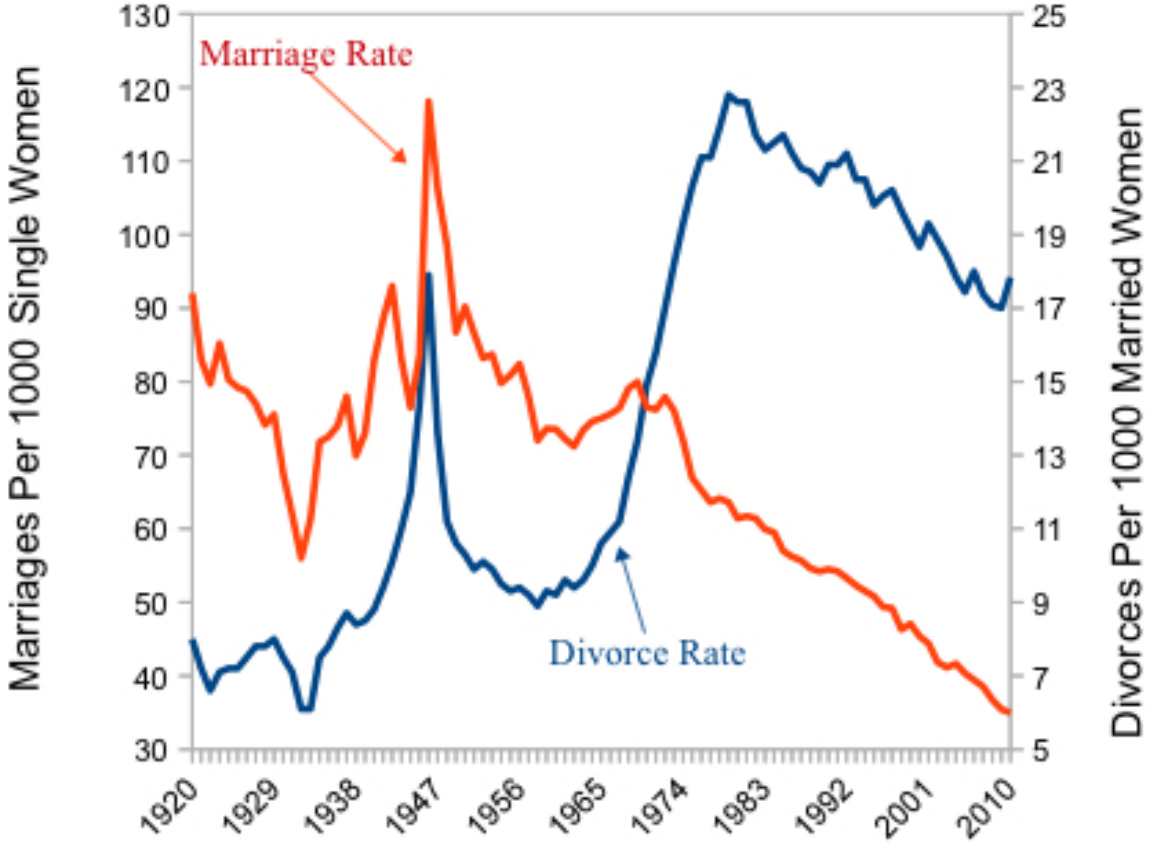
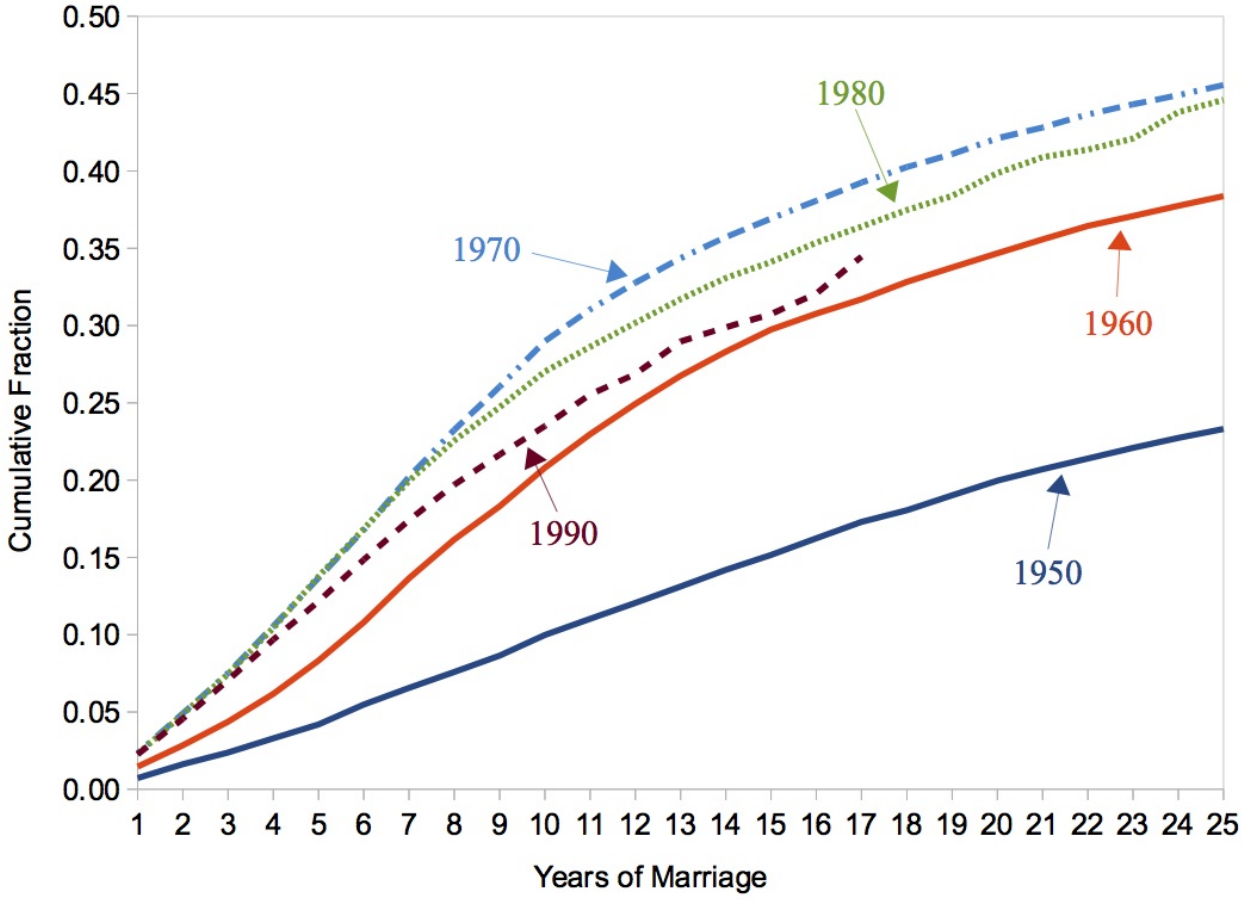
Marriage and Divorce Rates over Time

- Marriage & Divorce rates have been decreasing (recently).
- Tend to move together.



Source: Authors Calculation from Historical Statistics of the U.S. / National Vital Statistics / American Community Survey (ACS) More On Marriage U-Shape

Decreasing Divorce



Source: First Marriages. Whites. Authors Calculation from Combined 2001, 2004 & 2008 SIPP

HS or less HS or more

"The Gray Divorce Revolution"

- Overall decline in divorce rates, but...
- ... divorce rates increased among the elderly:

Divorce Rates per 1000 married men	**1980**	**2007**
20-49	33.25	26.39
50-59	7.69	15.86
60 plus	2.47	6.71

Source: Historical Statistics of the U.S. / American Community Survey (ACS)

Marriage rates by age

More cross-state marriage/divorce

More LFP and Divorce

Method

- Build an equilibrium search model of the marriage market with:
  - endogenous marriage/divorce.
  - endogenous fertility.
- Calibrate to the US economy in 2005.
- Change earnings volatility to its 1970 level:
  - Compare model's change of marriage and divorce with the data.
  - Preview of (preliminary) results:
    - 17% of change in marriage rates.
    - 35% of change in divorce.
    - 9% of change in elderly divorce rates.

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Literature

- Consumption Commitments:
  - Postlewaite, Samuelson and Silverman (2008), Chetty and Szeidl (2007)
- Volatility:
  - Heathcote, Perri and Violante (2010), etc.
- Families/Marriage/Risk
  - Cubeddu & Rios-Rull (1997, 2002)
  - Sommers (2012)
  - Santos & Weiss (2012)
  - Gould & Passerman (2003)
- Divorce
  - Rotz (2011)
  - Ne'eman, Newman & Olivetti (2008)

Model - Basics

- Demographics:
  - Two genders.
  - Life-cycle - 4 ages: 20s, 30s, 40/50s, 60+ (retired).
  - Single and married individuals.
- Stochastic labor income.
- Risk-Free Assets.
- Married households differ:
  - Economies of scale in consumption (public good).
  - Fertility choice (extensive margin).
  - Extensive margin female LFP decision.
  - Intrahousehold bargaining.
  - Divorce choice (unilateral).

Labor Income

- Individual income: $\underbrace{w}_{\text{Shock}} \quad \underbrace{\phi_g}_{\text{Gender Gap}} \quad \underbrace{f(t)}_{\text{Age Profile}}$

- Shock process w is
  - Dependent on marital status.
  - Correlated between spouses.

Shock Equations

Preferences

- Consumption:

$$u(c) = \frac{c^{1-\lambda}}{1-\lambda}$$

- Expenditures on children:

$$v(c_k) = \alpha \frac{(c_k - \underline{c_k})^{1-\lambda}}{1-\lambda}$$

- Disutility of working:

$$\psi \in \{\psi_l, \psi_h\}, \text{ where } \psi_l < \psi_h$$

- If a wife works when there is a kid in the household:
  - Extra disutility ψ_k

Preferences II

- Marital bliss shocks b:
  - Initial Draw: $b \sim N(\mu_b, \sigma_b^2)$
  - Updates: $b = \delta_b b_{-1} + \varepsilon_b$, $\varepsilon_b \sim N(0, \sigma_{bb}^2)$
  - Married parents get deterministic $\bar{b_k}$
- Extra transitory utility during marriage:

$$\gamma_d \sim N\left(0, \sigma_{\gamma_d}^2\right)$$

- Shock to desirability of kids (timing of fertility):

$$\gamma_f \sim N\left(0, \sigma_{\gamma_f}^2\right)$$

- Utility cost of divorce - $\omega(k)$

Budget Sets

- Singles:

$$\begin{cases} c + a' = w\phi_g f(t) + (1+r)a & \text{if childless} \\ c + a' + c_k = w\phi_g f(t) + (1+r)a + \tau & \text{if mother} \\ c + a' = wf(t) + (1+r)a - \tau & \text{if father} \end{cases}$$

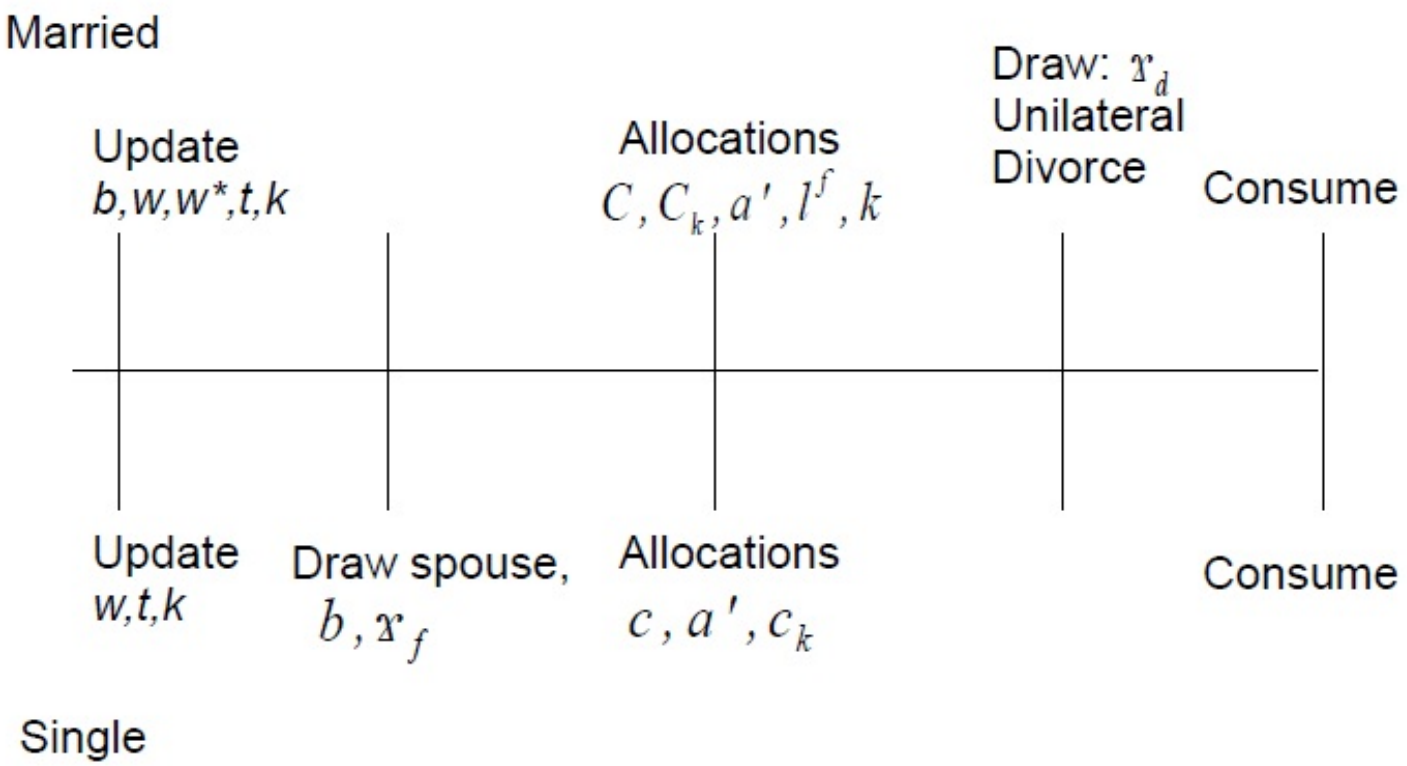
- Married:

$$\frac{c}{1+\xi} + a' + c_k = w_1 f(t) + I^f w_2 \phi f(t) + (1+r)a - \tau,$$

  - where $I^f \in \{0,1\}$ (LFP status).

- Exogenous government supplied consumption floor $\underline{c}$.

Timing


Married
Update
b,w,w*,t,k
Allocations
C, C_k, a', l^f, k
Draw: $\mathfrak{x}_d$
Unilateral
Divorce
Consume
Update
w,t,k
Draw spouse,
$b, \mathfrak{x}_f$
Allocations
c, a', c_k
Consume
Single

Decision Making - Singles

- State vector (w, a, τ, k, ψ, t):
  - w = wage shock
  - a = asset holdings
  - τ = child support transfers
  - k = whether there is a child in the household
  - ψ = disutility level of working
  - t = age
- Value function (men and childless women):

$$V_g^s(w, a, \tau, k, \psi, t) = \max_{c, a'} u(c) + \beta E_{w', t'} \underbrace{B_g}_{\text{Bachelor Phase}} (w', a', \tau, k, \psi, t')$$

- Value function (women with children):

$$V_2^s(w, a, \tau, k, \psi, t) = \max_{c, c_k, a'} u(c) + v(c_k) + \beta E_{w', t'} \underbrace{B_2}_{\text{Bachelor Phase}} (w', a', \tau, k, \psi, t')$$

Decision Making - Marriage

- Decision is bilateral. A marriage occurs iff

$$V_g^m(w, w^*, a + a^*, \tau, k^*, b, \psi, t, \gamma_f) > V_g^s(w, a, \tau, \psi, t) \qquad (1)$$

  for both g.
- We summarize this decision in the following function:

$$J_m(w, w^*, a, a^*, \tau, \tau^*, k^*, b, \psi, t) = \begin{cases} 1 & \text{if (1) holds for all } g \\ 0 & \text{otherwise} \end{cases}$$

Decision Making - Marriage

- Value of the bachelor phase

$$B_g(w, a, \tau, k, \psi, t) = \int\int\int \{\underbrace{J_m\left[V_g^m\right]}_{\text{Marry}} + \underbrace{(1 - J_m)V_g^s}_{\text{Remain Single}}\}$$

$$dB(b)d\Gamma(\gamma_f)d\hat{\mathbf{S}}_{\mathbf{g}^*}(z^*)$$

Couples

- State vector $(w, w^*, a, \tau, k, b, \psi, t)$:
  - w = person's wage shock
  - w^* = spouse's wage shock
  - a = asset holdings
  - τ = child support transfers (from previous relationships)
  - k = whether or not one (or both) spouse have a child
  - b = marital bliss shock
  - ψ = disutility of wife working
  - t = age

Value Functions - Couples

The value function for a married individual read

$$V_g^m(w, w^*, a, \tau, k, b, \psi, \gamma_f, t) = \underbrace{u(\tilde{c})}_{\text{Consumption}} + \underbrace{v(\tilde{c}_k)k}_{\text{Kid's Consumption}} + \underbrace{b}_{\text{Marital Bliss}}$$

$$+ \underbrace{k\bar{b}_k}_{\text{Kid Bliss}} - \underbrace{\psi(k)\tilde{l}}_{\text{Disutility Work}} + \underbrace{\gamma_f \tilde{k}}_{\text{Utility New Child}} +$$

$$\beta E_{w', w^{*\prime}, b', \gamma_f', t'} V_g^m\left(w', w^{*\prime}, \tilde{a}, \tau, k, b', \psi, \gamma_f', t'\right)$$

$\tilde{Var}$- Determined by Nash Bargaining.

Decision Making - Couples

Dynamic bargaining problem:

$$V = \max_{c, c_k, a', l^f, k} \left(V_1^m(w, w^*, a, \tau, k, b, \psi, \gamma_f, t) - V_1^d(w, w^*, a, \tau, k, \psi, t) \right)^{\eta} \times$$
$$\left(V_2^m(w^*, w, a, \tau, k, b, \psi, \gamma_f, t) - V_2^d(w^*, w, a, \tau, k, \psi, t) \right)^{1-\eta}$$

subject to

$$c/(1+\xi) + a' + c_k = wf(t) + l^f \phi f(t) w^* + (1+r)a - \tau.$$

Decision Making - Divorce

Transfers- father to mother: $\tau = \chi w f_1(t)$.

Divorce value function:

- Female:

$$V_2^d(w^*, w, a, \tau, k, \psi, t) = V_2^s(w^*, a/2, \tau, k, \psi, t) - \omega_d(k)$$

- Male:

$$V_1^d(w, w^*, a, \tau, k, \psi, t) = V_1^s(w, a/2, \tau, t) - \omega_d(k) + \frac{1}{1-\beta(1-\eta(t))} v(\hat{c}_k(w^*, a/2, \tau, \psi, t))$$

$\hat{c}_k$: child expenditures made by ex-wife.
$\eta(t)$: rate child becomes adult.

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$\hat{c}_k$: child expenditures made by ex-wife.
$\eta(t)$: rate child becomes adult.

Decision Making - Divorce

- The decision is unilateral. A divorce will happen iff

$$V_g^m(w, w^*, a, \tau, k, b, \psi, \gamma_f, t) + \gamma_d < V_g^d(w, w^*, a, \tau, k, \psi, t)$$

for at least one g. γ_d is the transitory divorce shock.

Equilibrium

Definition

An equilibrium for this economy is a collection of value functions, policy functions, and distributions such that:

1. The value functions and the policy functions solve the agent's problem, given the distributions.

2. The policy functions give rise to the distributions.

Quantitative Exercise

- Calibrate to US economy in 2005.
  - Some parameters set a Priori.
  - Rest are chosen to match data.
  - Use Simulated Method of Moments.
- Feed in 1970s volatility.
- Compare changes in model marriage/divorce to the data changes.

Computation

Calibration - Parameters Set a Priori

Param.	Description	Value	Source
λ	CRRA	2	Standard
β	Discount	0.97	Standard
r	Interest Rate	1%	Risk-Free
ψ	Economies of Scale	0.70	OECD
η	Bargaining Weights	0.50	Standard
π_t	Probability of aging	20s, 30s, 40-50s, 60s	—
$f(t)$	Age income profile	—	ACS
ρ	Corr Spousal Income Shocks	0.25	Hyslop (2001)
$\underline{c}$	Consumption Floor	\$2640/year	Kaplan (2012)
χ	Child Support Obligations	0.25	Wisconsin Guideline

Estimation - Income Shocks

Income process in the data:

- PSID data: 1968-2009
- Separately estimate process for single and married individuals

Income Process

Gender wage gap in the data:

- ACS data: 2005. Observed Gender wage gap.
  - Control for: age, education.

Indirect Inference:

- Guess labor market parameter values
- Solve model
- Simulate model data
- Use same estimator on model data
- Get estimates from model data and actual data to match

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Calibration - Match Moments (2005)

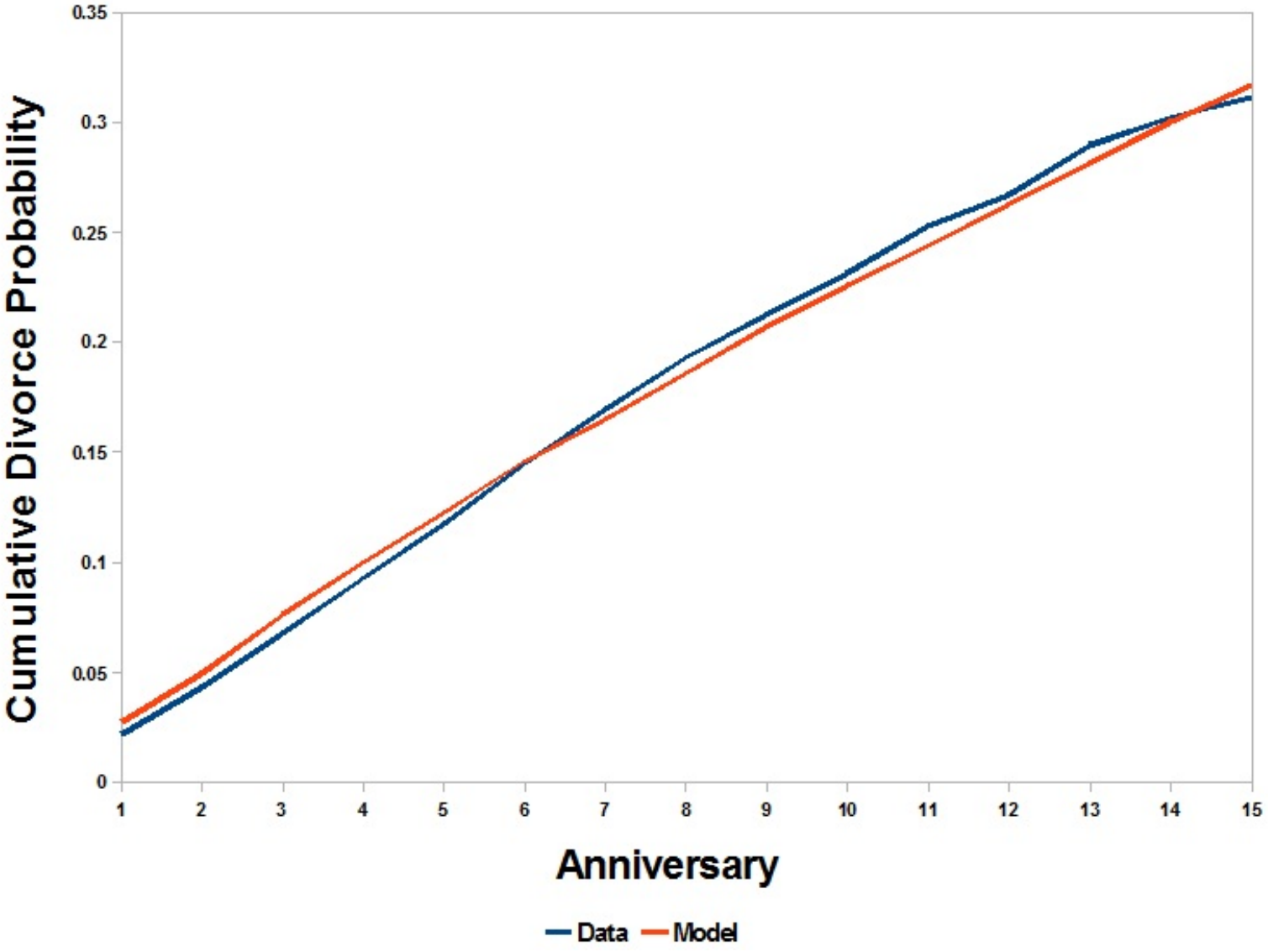
Param.	Description	Target
α	Utility Weight on Kids	Fraction HH expenditures on kids
$\underline{c_k}$	Subsistence level for kids	Rel. Fert Rates 20/30
μ_γ	Mean marital bliss shock	Marriage rates: lifecycle
σ_b	Dispersion of marital bliss draw	Marriage rates: lifecycle
δ_γ	Persistence of marital bliss shock	Divorce rates: marital tenure
σ_{γ_d}	Transitory Divorce Shock	Divorce rates: marital tenure
σ_{bb}	St. Dev of Mar Bliss Innov.	Divorce rates: lifecycle
$\omega_d(k)$	Utility Cost of Divorce	Rel. divorce rates with/without kids
$\bar{b}_k$	Flow utility of having kids	% Women have kid by 40
σ_f	St. Deviation of fertility bliss shock	Fertility rate of those w/o kids
ω_o	Divorce cost (old)	Elderly divorce rate
ψ_h	High cost of LFP	Married women's LFP
ψ_k	Extra cost of LFP with kids	Married women's LFP w/kids

Targets: 18

Parameters: 13

Including Labor Market: 23Targets, 18 Parameters

Model Fit - Divorce by Marital Tenure



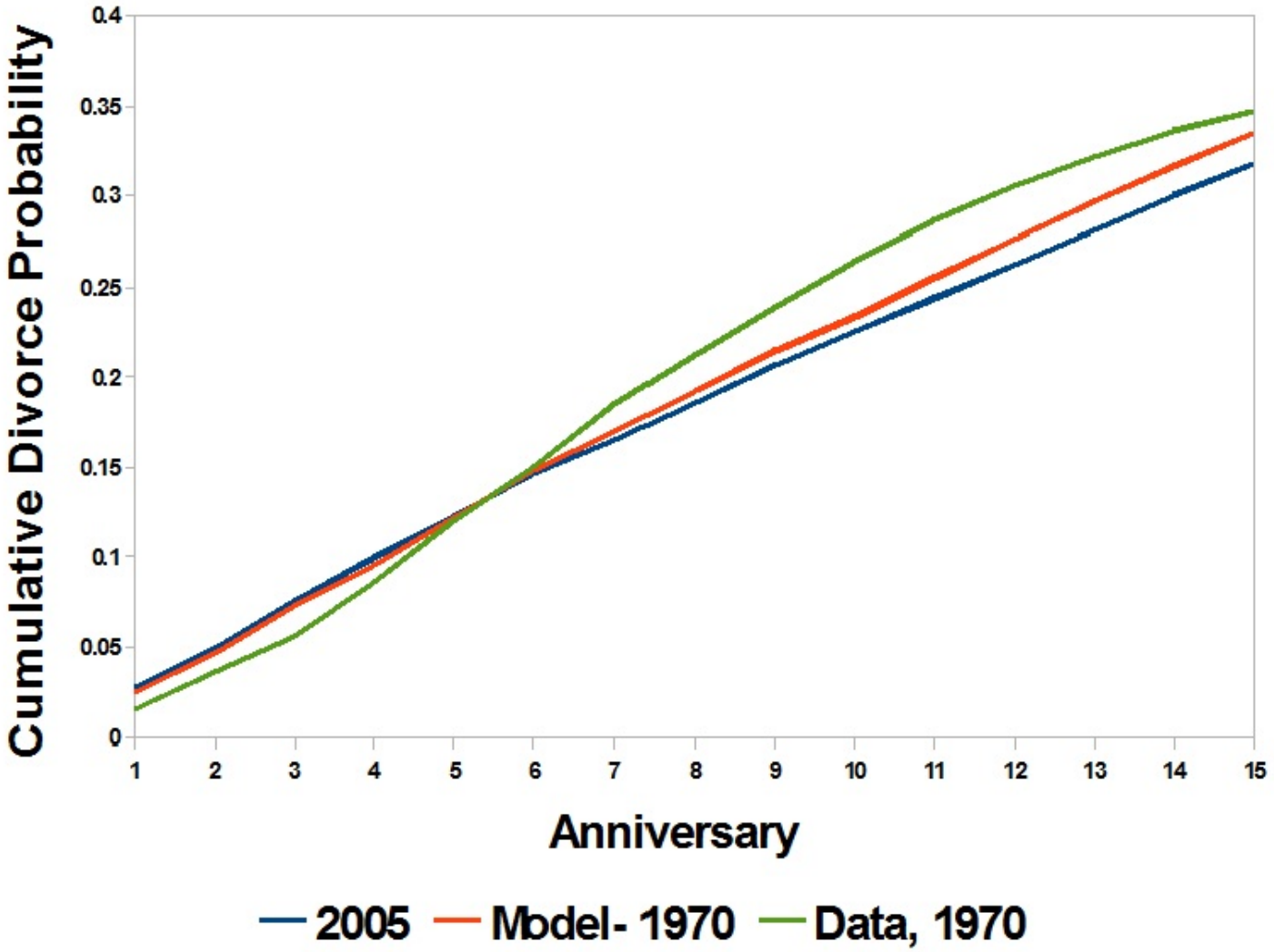
Model Fit- 2005

Moment	Data	Model
Fraction HH expenditures on kids	45%	48%
Ratio 20/30 adj fert rates	0.39	0.46
Marriage rates: 20s/30s	8.29%	16.56%
Marriage rates: 40s/50s	3.36%	2.17%
Marriage rates: 60s	0.90%	0.71%
Divorce rates: 20s	4.08%	4.33%
Divorce rates: 30s	3.21%	2.84%
Divorce rates: 40s/50s	1.94%	1.90%
Divorce rates: 60s	0.73%	0.27%
Rel. divorce rates with/without kids	0.68	0.54
% Women have kid by 40	77.3%	82%
Fertility rate w/o kids	28.1%	35%
Married women's LFP w/o kids	73%	75%
Married women's LFP w/kids	67%	59%
Single Persistence	0.929	0.922
Single Variance	0.027	0.024
Married Persistence	0.976	0.976
Married Variance	0.016	0.015

Parameters

Counterfactual- Cumulative Divorce Rate

- Model explains approx. 35% of the change in the divorce profile.



Percent Explained

Counterfactual - Rates

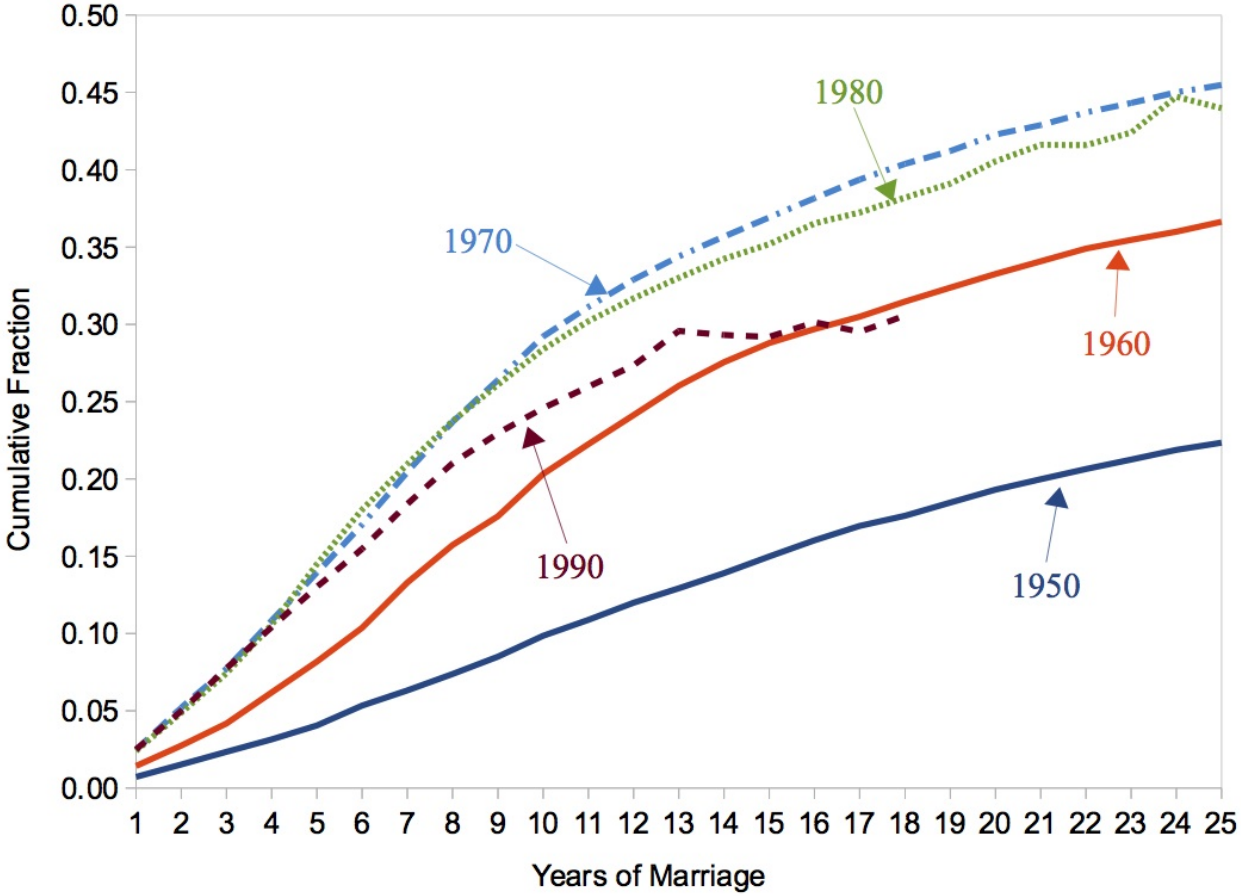
Feed into the model the 1970 level of income volatility

Moment of Interest	**% Change Data**	**% Change Model**	**Model/Data Change**
Marriage Rates	$\downarrow$47%	$\downarrow$8%	17%
Divorce Rates (60s)	$\uparrow$ 300%	$\uparrow$ 16%	9%

Conclusion

- Families are sensitive to labor market risk. Volatility leads to:
  - Less marriage.
  - Less divorce.
  - More elderly divorce.
  - These joint movements are hard to get from other mechanisms.
- Nash Bargaining with marriage/divorce/fertility/savings:
  - First (we believe) to have it all.
  - Interesting computational issues.
- (Continuing work) can study savings/labor supply/child welfare.

% of Marriages that Ended in Divorce (HS or less)



Source: First Marriages. Whites. Authors Calculation from Combined 2001, 2004 & 2008 SIPP

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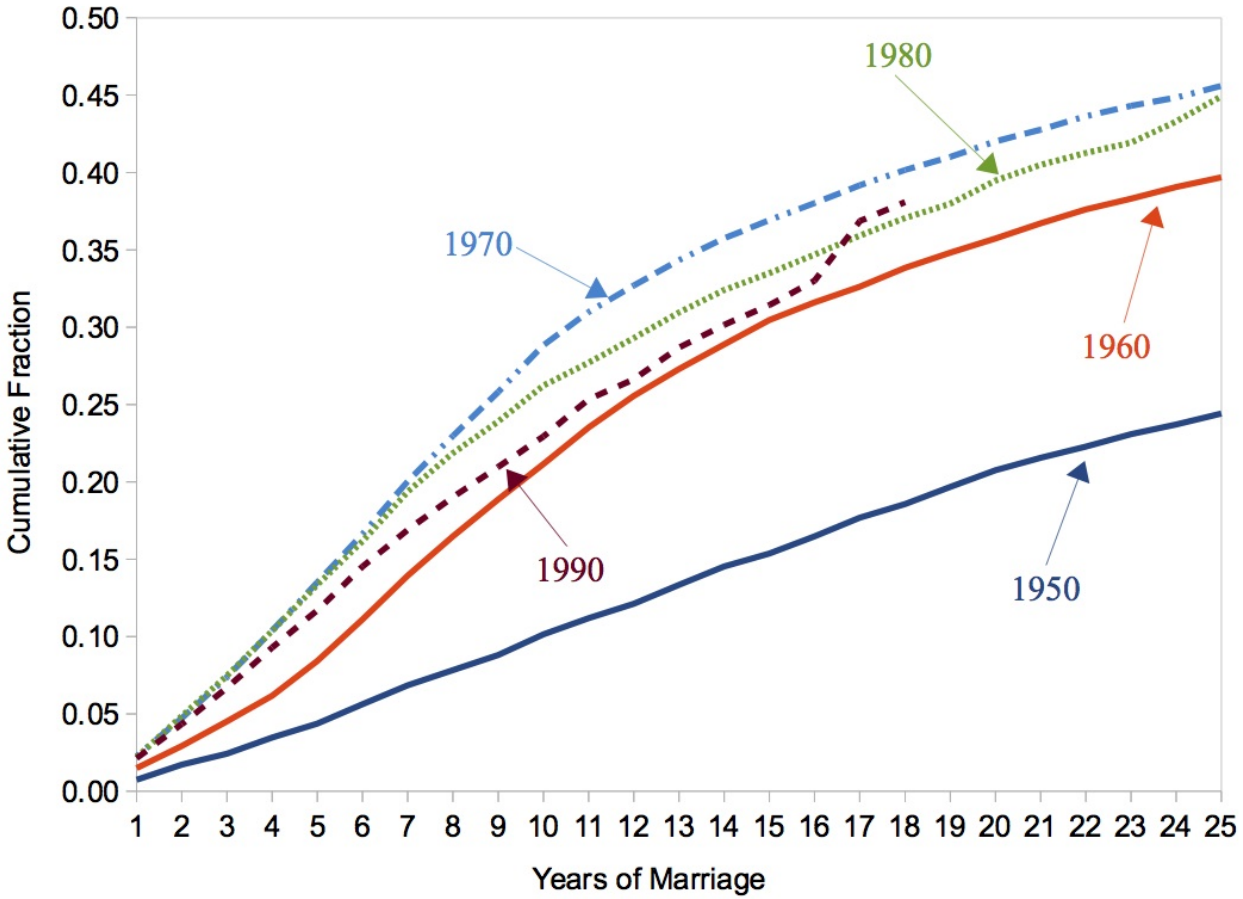
% of Marriages that Ended in Divorce (more than HS)



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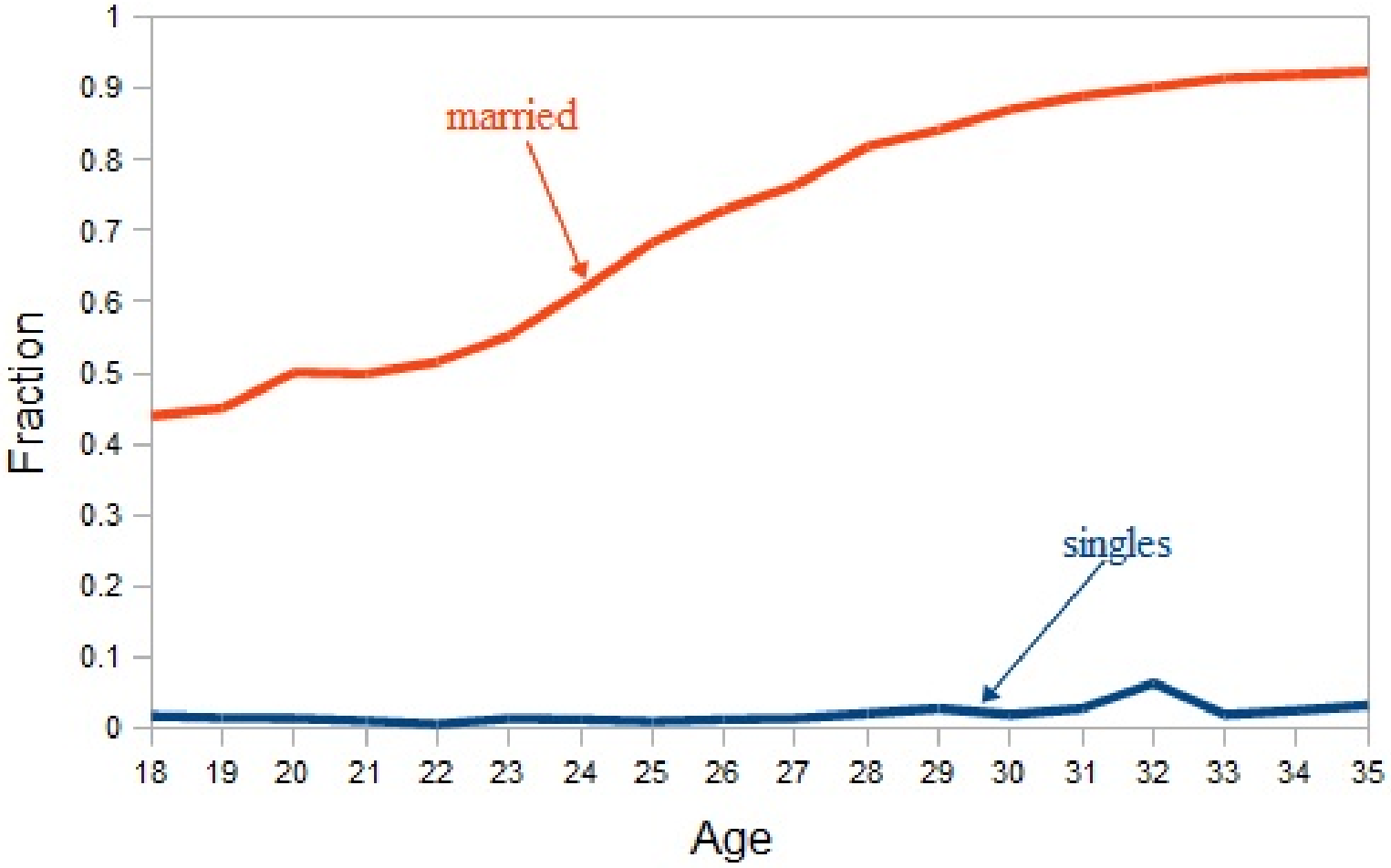
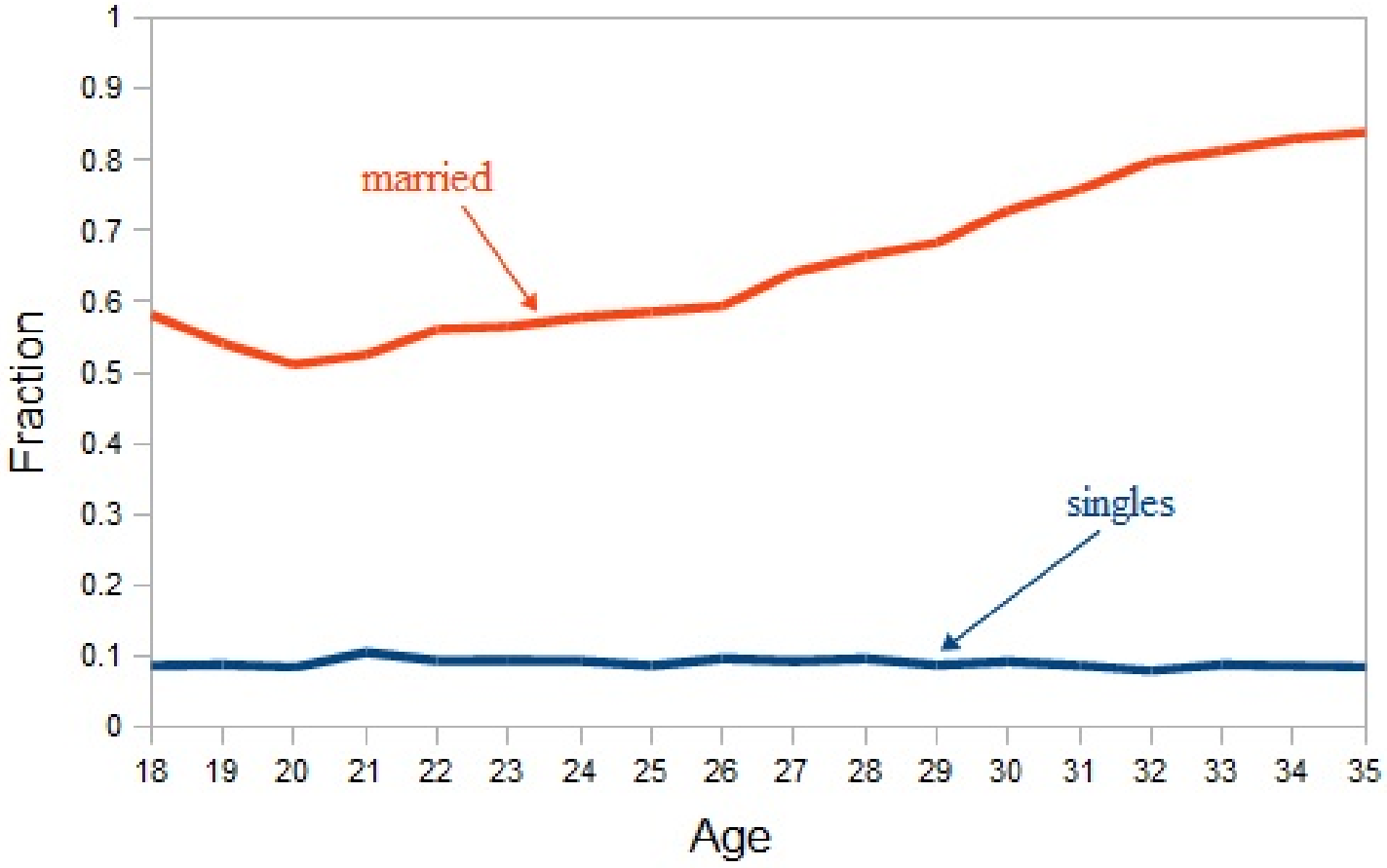
Presence of Children by Marital Status and Age - 1970



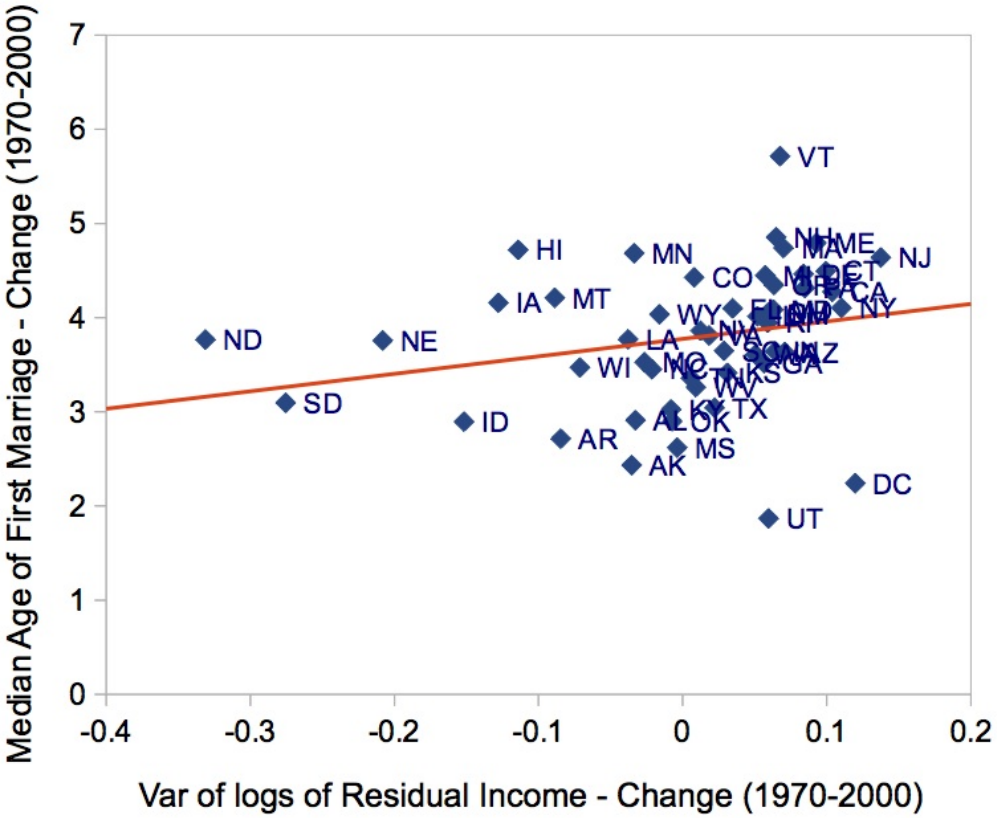
Source: US Census

Presence of Children by Marital Status and Age - 2000



Source: US Census

Cross-state: Marriage and Residual Income Inequality



Source: Authors Calculation, Census and National Vital Statistics

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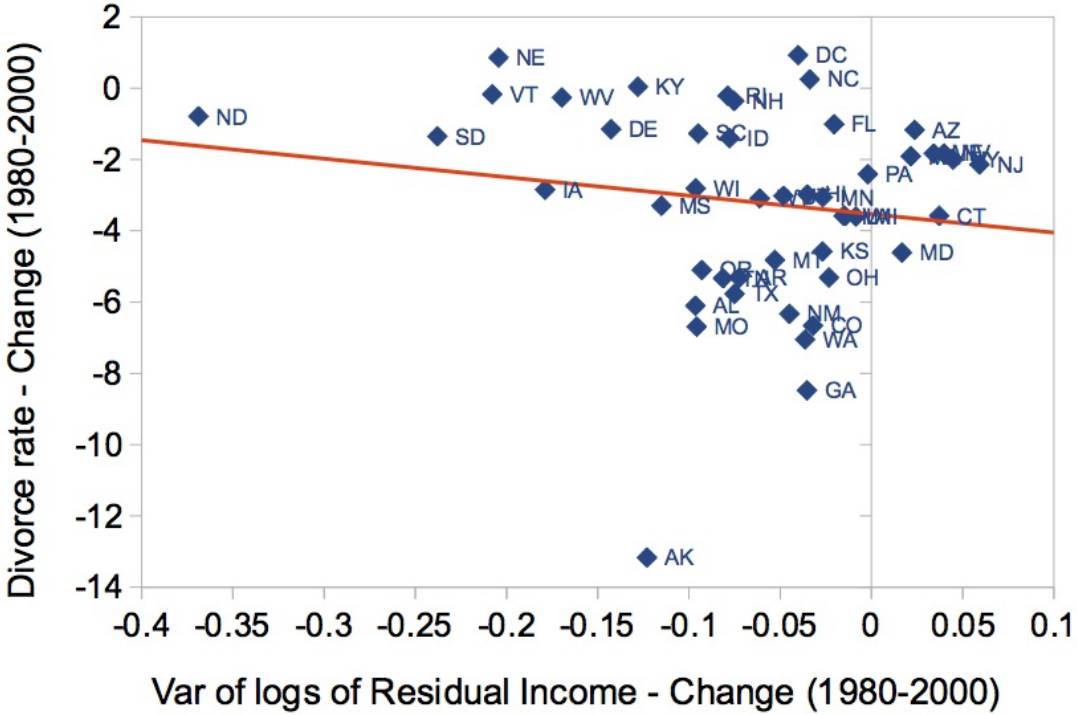
Cross-state: Divorce and Residual Income Inequality



Source: Authors Calculation, Census and National Vital Statistics

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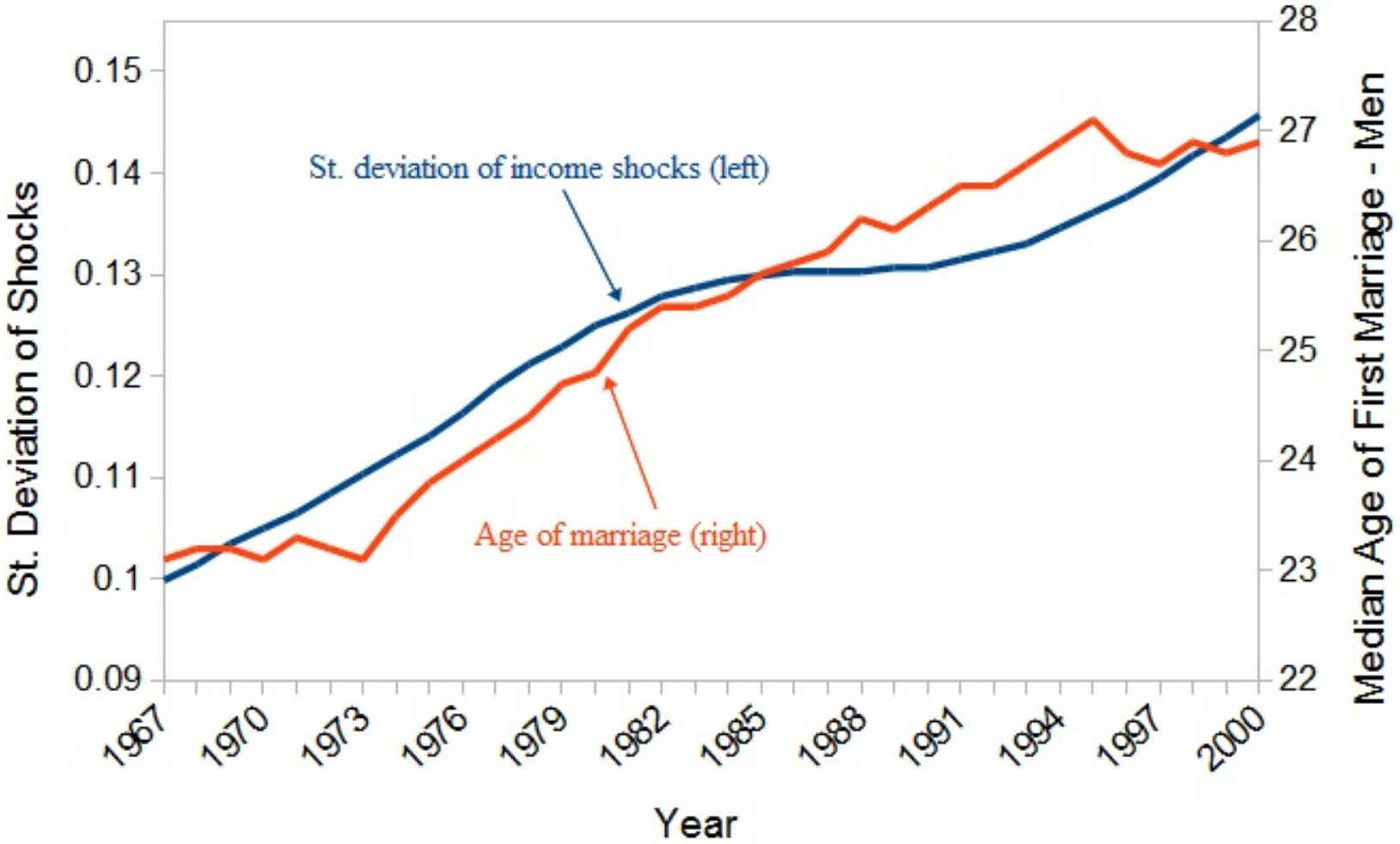
Earnings Volatility and Age at Marriage



Source: Authors Calculation from PSID & HSUS

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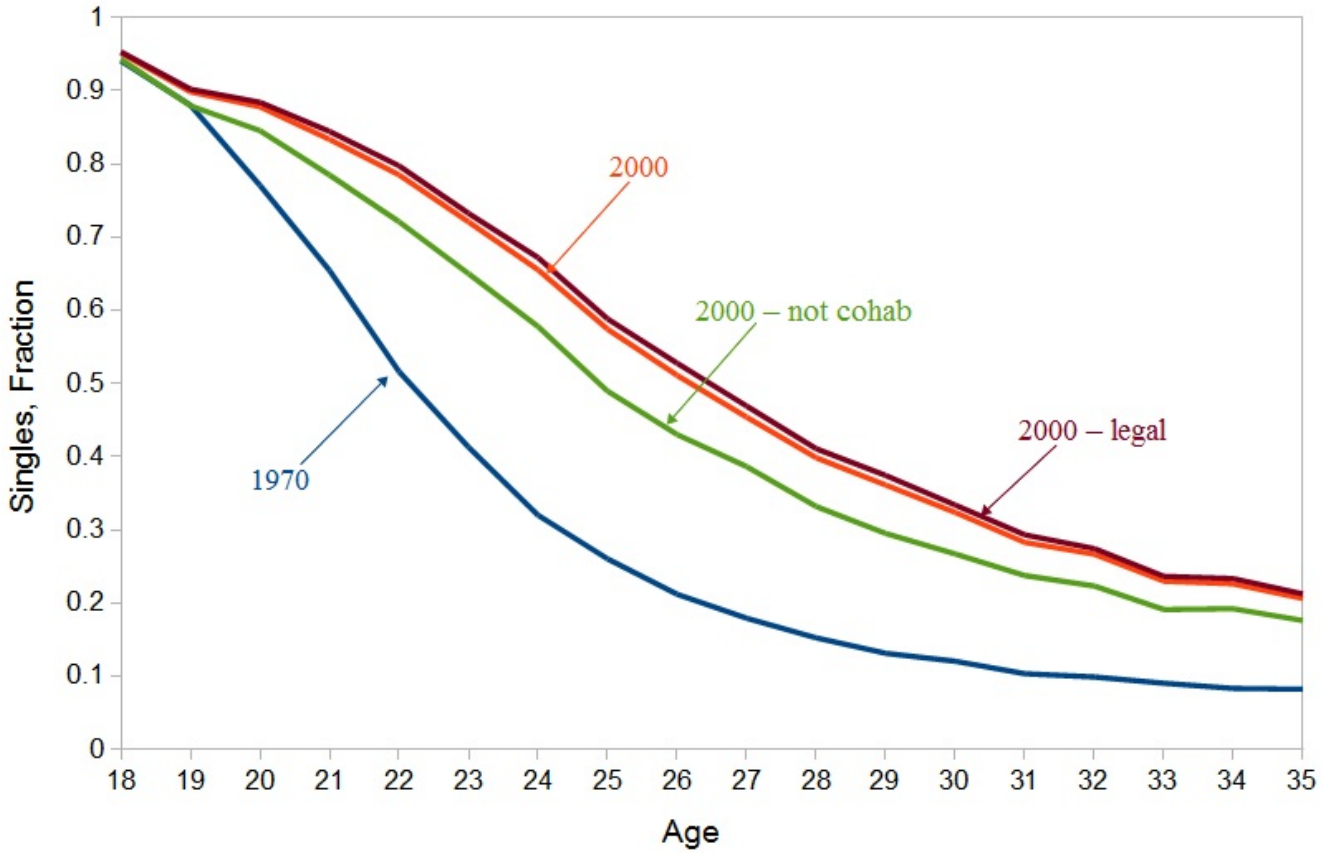
Singlehood, Marriage, and Cohabitation



Source: US Census

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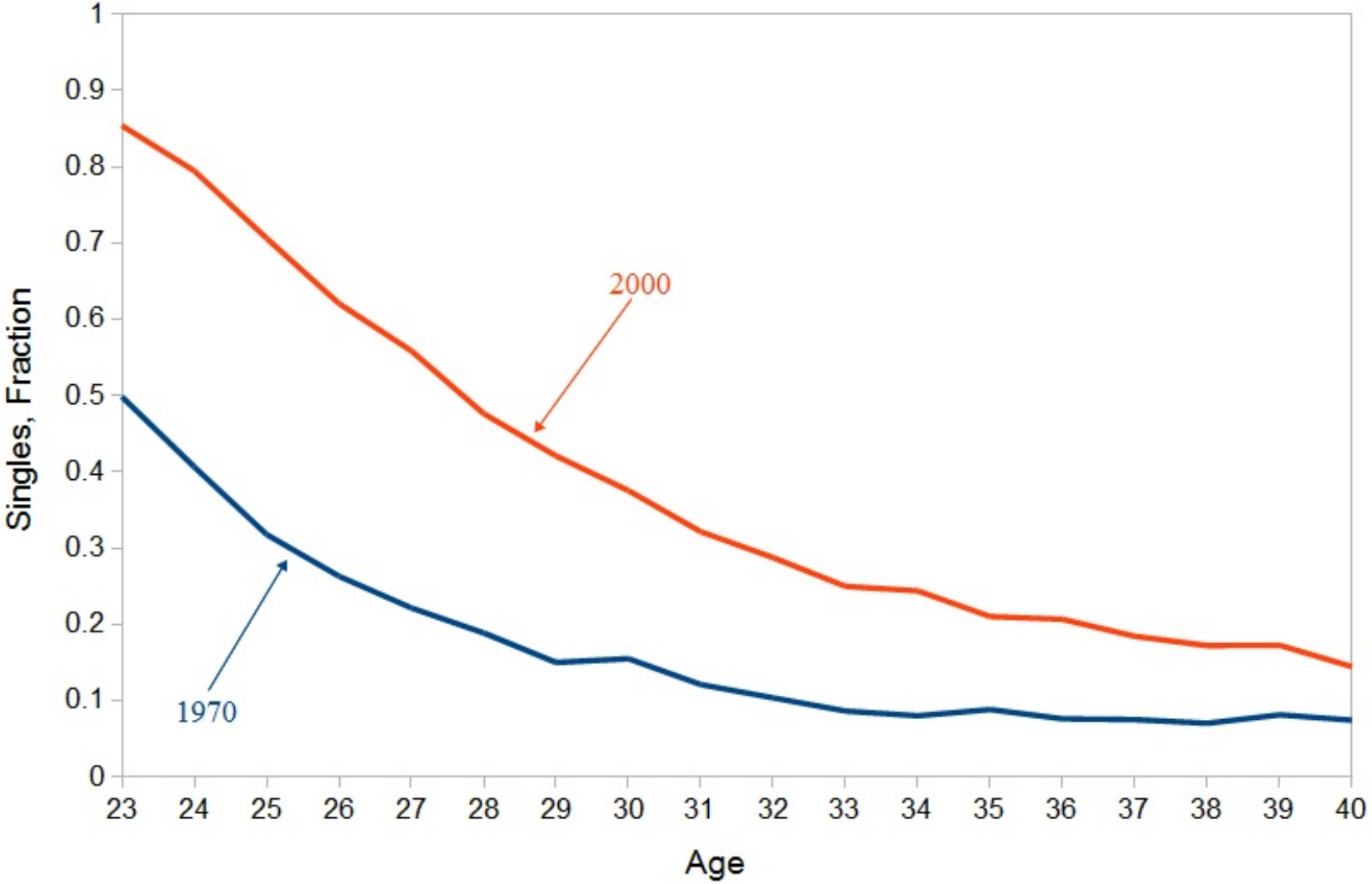
Marriage, College



Source: US Census

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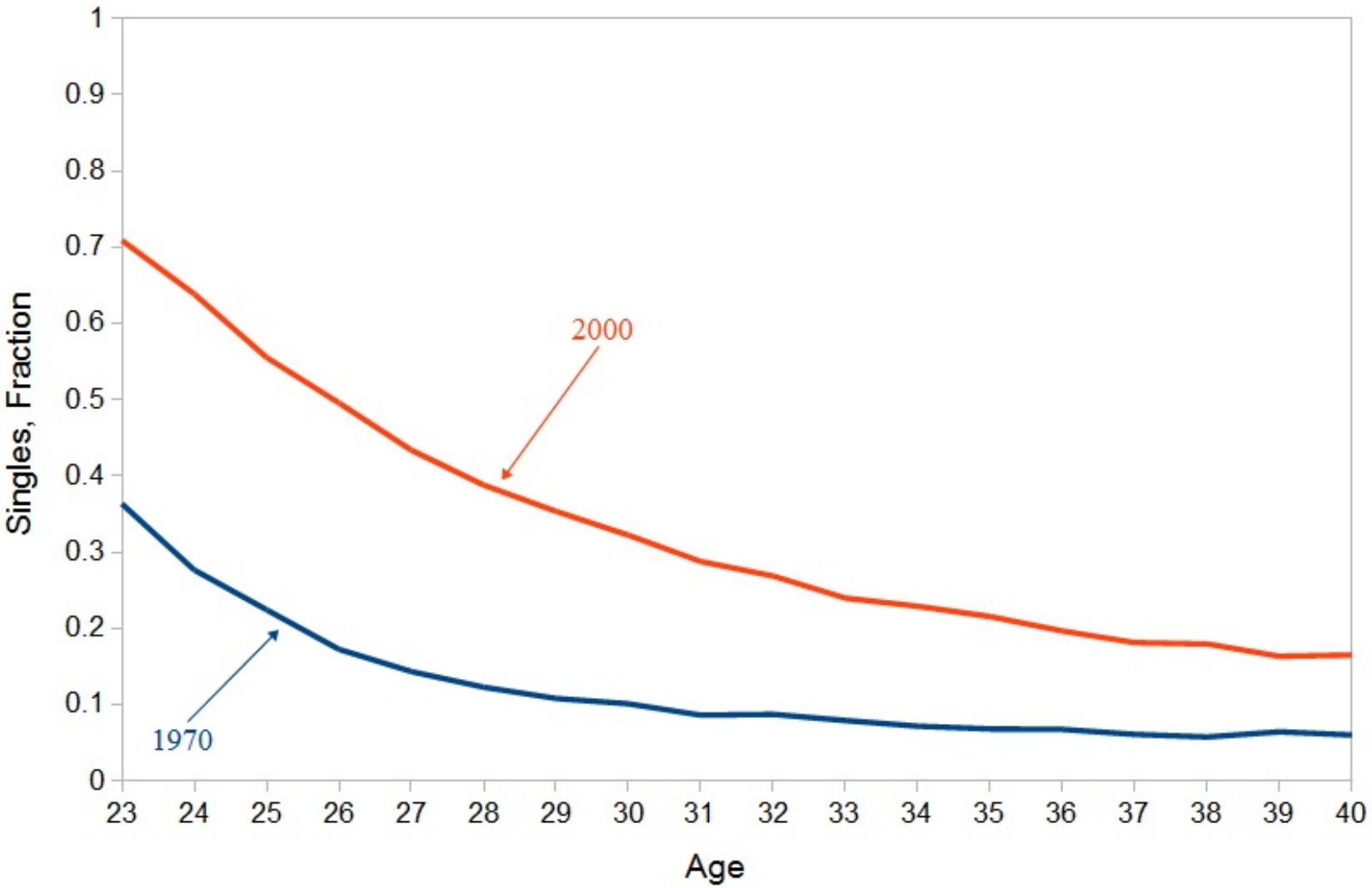
Marriage, No College



Source: US Census

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Marriage Rates, by age

Marriage Rates per 1000 single men	1970	1980	2007
20-49	201.56	127.22	59.58
50-59	59.52	46.38	29.74
60 plus	22.21	22.4	15.55

Source: Vital Statistics and ACS

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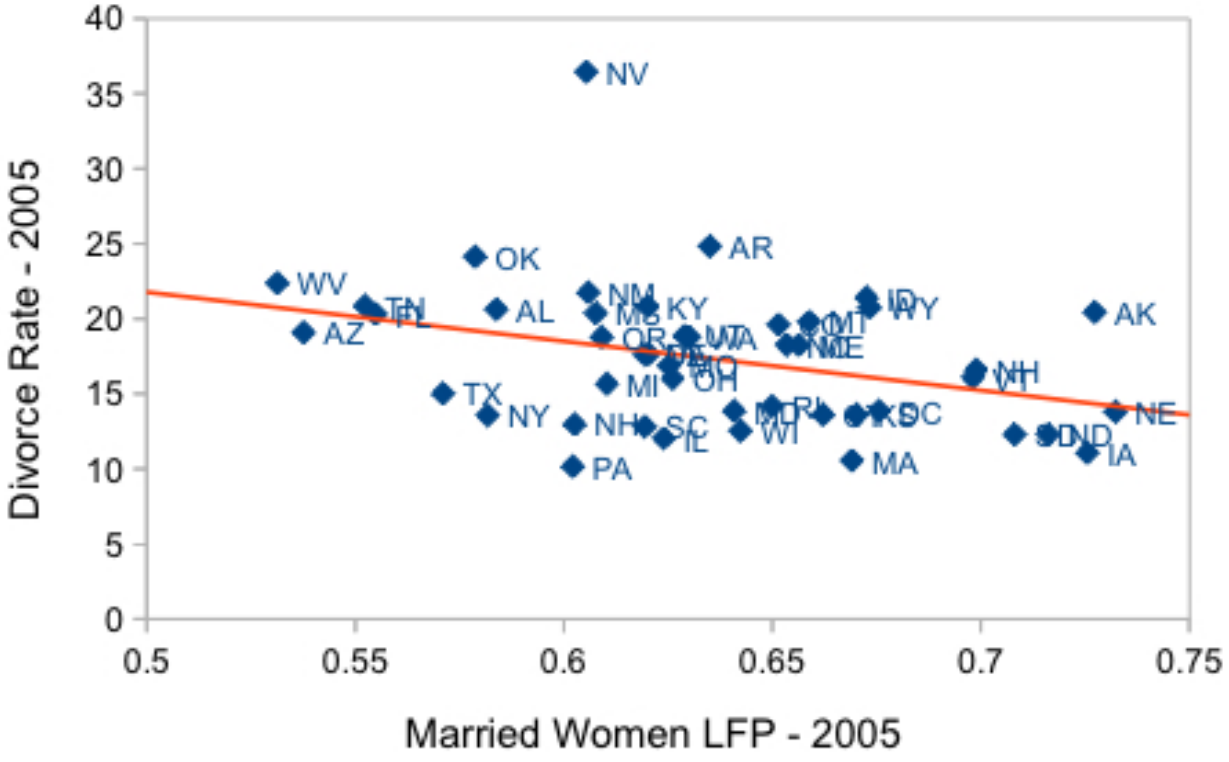
Divorce and Married Women LFP



Source: Authors Calculation from National Vital Statistics & CPS

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Income Process Equations

- Idiosyncratic shocks - singles:

$$\begin{aligned} \ln w_s &= \delta_s \ln w_{s,-1} + \eta_s \\ \eta_s &\sim N(0, \sigma^2_{\eta,s}) \end{aligned}$$

- Idiosyncratic shocks - married:

$$\begin{aligned} \ln \vec{w}_m &= \delta_m \ln \vec{w}_{m,-1} + \vec{\eta}_m \\ \vec{\eta}_m &\sim N\left(0, \begin{bmatrix} \sigma^2_{\eta,m} & \rho \\ \rho & \sigma^2_{\eta,m} \end{bmatrix}\right) \end{aligned}$$

ρ: correlation between spouse's persistent shocks

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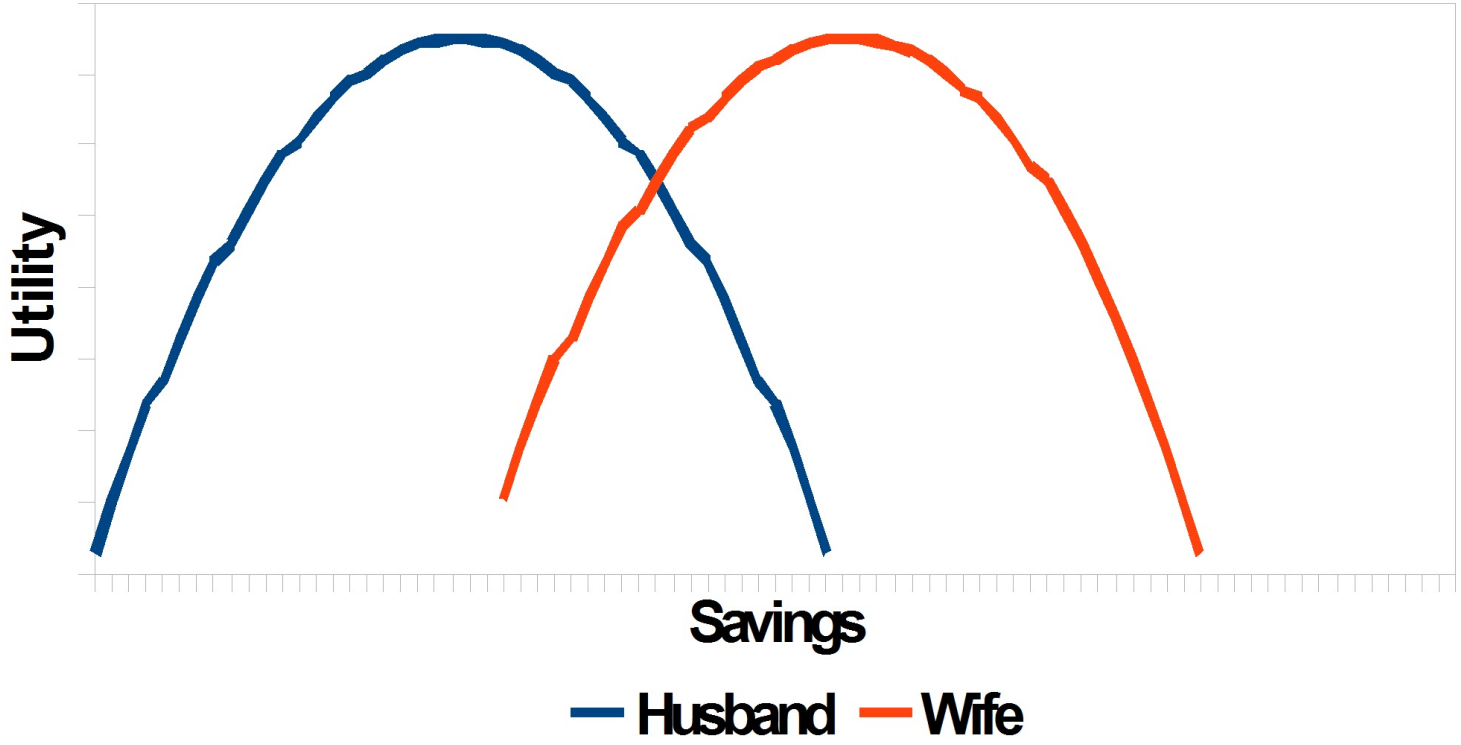
Computation- Married Problem

- Convert the Nash Bargaining problem to an equivalent Pareto problem with weight $\theta_p = \frac{V_w^m - V_w^d}{(V_w^{mc} - V_w^d) + (V_h^{mc} - V_h^d)}$.

Figure:



Computation- Problem

- Problem: Local maxima
- Small changes in θ_p will lead to changes in allocations and thus individiual value functions not converging.

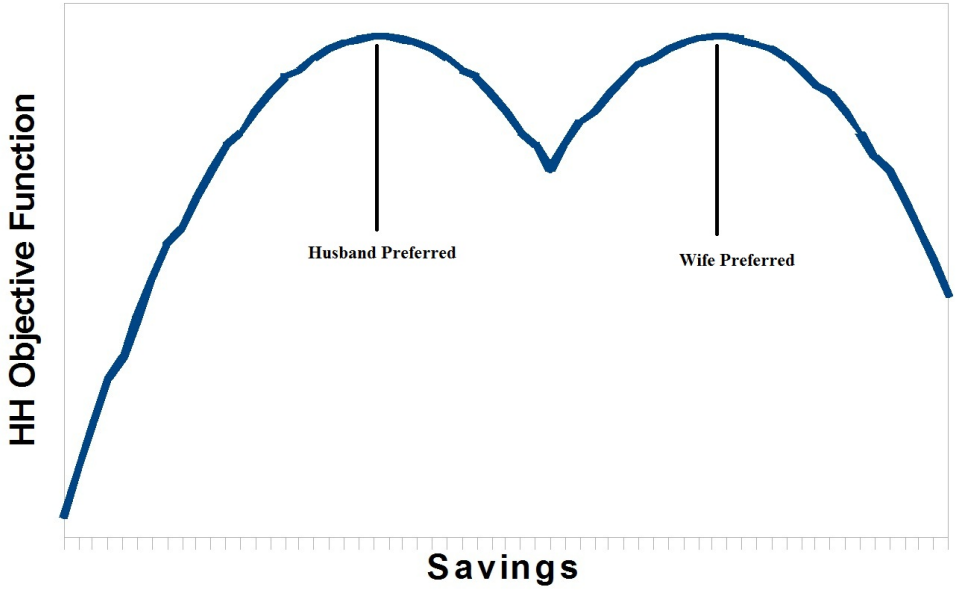


Figure:

Computation- Solution

- Use Quantal Response Equilibrium (QRE) over discrete choices.
- QRE:
  - Agents "play" all feasible allocations with positive probability
  - Probabilities are given by:

$$P(i) = \frac{e^{\chi V(i)}}{\sum_i e^{\chi V(i)}},$$

    where i represents each allocation and $V(i)$ is the value for the household of this particular allocation
  - As $\chi \to \infty$, QRE converges to playing the best response with probability one.

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Parameters

Param.	Description	Value
α	Utility Weight on Kids	1.62
$\underline{c_k}$	Subsistence level for kids	0.73
μ_γ	Mean marital bliss shock	-15.3
σ_b	Dispersion of marital bliss draw	10.57
δ_γ	Persistence of marital bliss shock	.90
σ_{γ_d}	Transitory Divorce Shock	25.57
σ_{bb}	St. Dev of Mar Bliss Innov.	1.76
ω_d	Utility Cost of Divorce	100
$\bar{b}_k$	Flow utility of having kids	15.15
σ_f	St. Deviation of fertility bliss shock	109.40
$\omega_{d,o}$	Divorce cost (old)	10.45
ψ_h	High cost of LFP	0.24
ψ_k	Extra cost of LFP with kids	1.60

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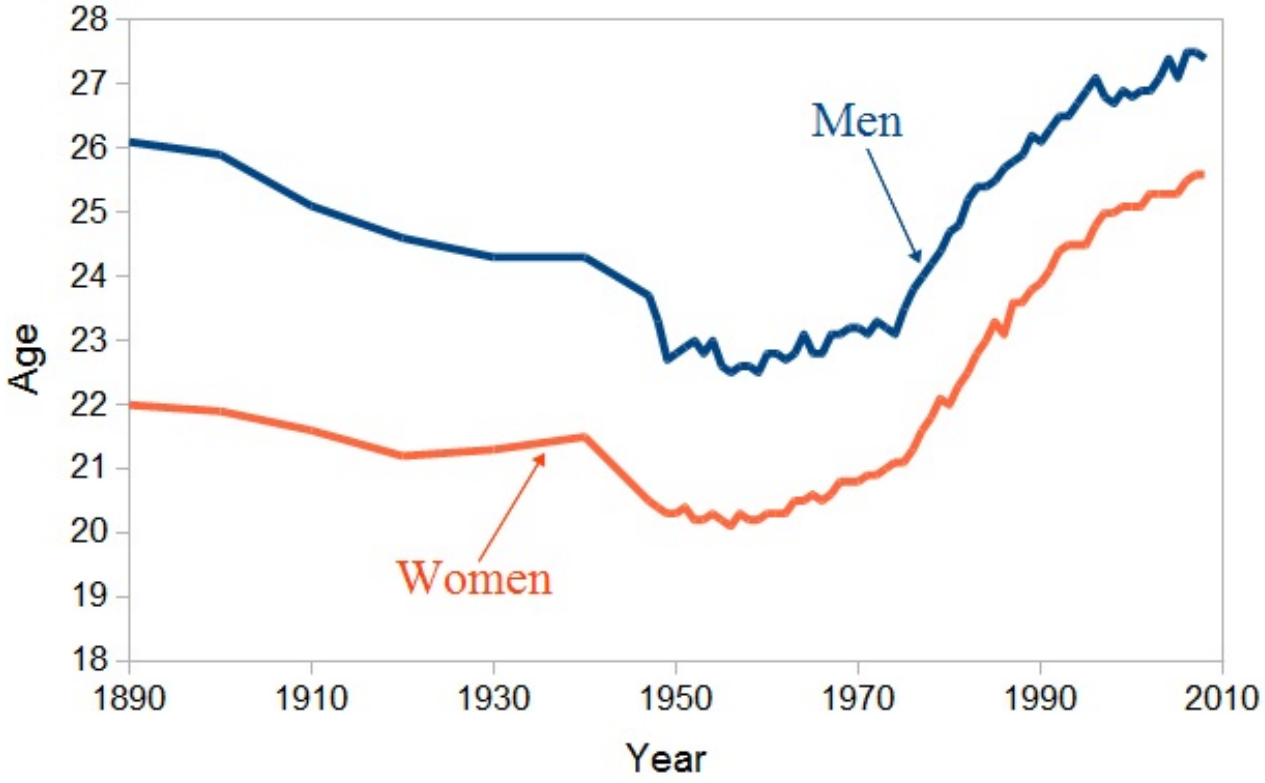
Marriage U-Shape



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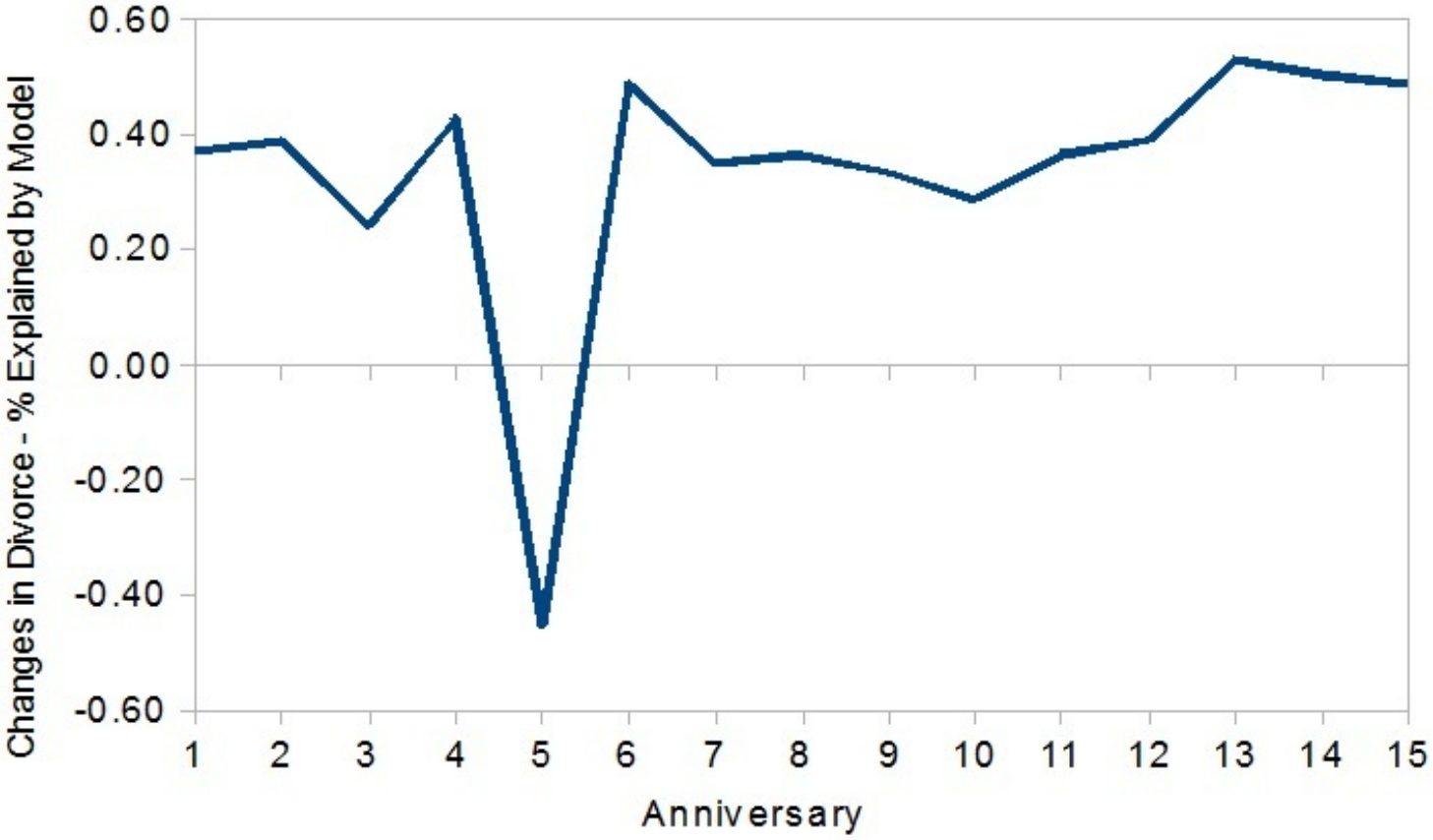
Divorce Change Explained



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Idiosyncratic Income Shocks

Using the residuals, estimate:

$$
\begin{aligned}
\ln w_t &= \alpha + \underbrace{z_t}_{\text{persistent}} + \underbrace{\varepsilon_t}_{\text{transitory}} \\
z_t &= \delta z_{t-1} + \eta_t \\
\eta_t &\sim N(0, \sigma^2_{\eta,t}) \\
\varepsilon_t &\sim N(0, \sigma^2_{\varepsilon,t}).
\end{aligned}
$$

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